

1. Foundational Concepts

- a. In a few sentences, describe the differences between affordances and signifiers.

[6 marks]

- b. Which three of Nielsen's heuristics would be of most use to a designer creating a new shopping application, and why?

Important: there is no 'right' answer here. We want to understand how you recall and apply the terminology.

[6 marks]

- c. What is a conceptual model? Give an example of one that you think is useful.

[6 marks]

- d. What is thematic analysis and when could you use it?

[7 marks]

2. Empathise – envision the needs and challenges of your users.

- a.** You have been hired by GolfClub, a social golfing platform for golfers wanting to meetup and play golf. Your aim is to improve the user experience for golfers by better understanding their needs.

You are to interview 15 future users from your population. Assume you have already ethics, consent and have introduced the interviewee to the topic.

- i. Briefly describe your recruitment process and target demographic.
- ii. Draft an interview guide you could use, including 8-10 key questions you would ask.
- iii. Outline the key areas these questions will be covering – i.e., what aspects of the users' experience are you trying to learn more about and why.

[17 marks]

- b.** Sketch an empathy map diagram and fill it out with 3 observations for each quadrant that you could imagine emerging from the interviews. Do not worry about what exactly your imagined observations say; we are trying to see how you are able to think about the aims of the empathy map process, rather than how well you imagine what 'real' users would say.

[8 marks]

3. Prototype

- a. You are working for a sporting company which is famous for its running shoes. The results of your empathise phase have indicated that there is a need for runners to have more detailed metrics of their performance. Your ideate phase indicated that there is potential in a smart shoe with a simple visual display and on-board sensors to help capture detailed running information.
 - i. For the first prototype, your boss has asked you to design it in high-fidelity. Write an argument to your boss explaining why it is a bad idea.
 - ii. Describe a Wizard of Oz prototype to test the minimum viable product. You may show simple sketches if you like.
 - iii. What other prototyping methods would be most useful to explore this idea? Use theory to explain your ideas.

[20 marks]

4. Evaluate

Your local doctor has developed a new app to help its customers with booking appointments. They want to know if their new prototype works better than their previous email service.

- a. Outline what would be useful evaluation metrics for your system.
 - i. Identify 3 qualitative or quantitative indicators that you could use to show your prototype is useful compared to the previous service.
 - ii. Explain why you selected these indicators, and how they might be measured.

[15 marks]

- b. Outline two evaluation approaches you could use to answer the question in the context of this design problem. One should be more 'in the wild', and the other should be more controlled.
 - i. Describe the rationale for each choice.
 - ii. What questions would **not** be answered by each approach?

[15 marks]